

Xin Li

AI Researcher / Ph.D. Student

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Profile

I am an AI researcher specializing in the intersection of large language models, robotics, and wireless communications. My research focuses on leveraging LLMs to automate mathematically rigorous tasks in engineering workflows, with applications in wireless system design, robotic perception, and SLAM. With extensive industry experience at Microsoft Research Asia, MEGVII, and Gausium Robotics, I have led teams in developing cutting-edge perception systems and contributed to top-tier venues including IEEE RA-L, IROS, ACL, and ICASSP. My work bridges theoretical advances with practical engineering solutions, achieving 100+ citations and recognition in international competitions.

Education

- 2025–Present **Ph.D. in Electrical and Electronic Engineering**, *Nanyang Technological University (NTU)*, Singapore
Advisor: Prof. Chau Yuen (IEEE Fellow)
Focus: Large Language Models for Engineering Automation
- 2018–2021 **M.Eng. in Electronics and Communication Engineering**, *Peking University*, Beijing, China
Thesis: Leveraging Geometric Regularities for Robust Visual-Inertial Odometry
- 2014–2018 **B.Eng. in Measurement and Control Technology**, *Northeastern University*, Qinhuangdao, China

Selected Publications

(* indicates equal contribution; see Google Scholar for complete list)

Large Language Models & AI

- ACL 2025 Findings **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**

Xin Li, Mengbing Liu, Li Wei, Jiancheng An, Merouane Debbah, Chau Yuen
Findings of the Association for Computational Linguistics (ACL), 2025
[Project Page] [Code]

- ML4RS@ICLR 2025 **Onboard Terrain Classification via Stacked Intelligent Metasurface-Diffractive Deep Neural Networks**

Mengbing Liu, Xin Li, Jiancheng An, Chau Yuen
ICLR Workshop on Machine Learning for Remote Sensing, 2025
[Project Page]

Wireless Communications & Signal Processing

- ICASSP 2025 **TransPathNet: A Two-Stage Framework for Indoor Radio Map Prediction**
Xin Li, Ran Liu, Saihua Xu, Sirajudeen Gulam Razul, Chau Yuen
IEEE International Conference on Acoustics, Speech and Signal Processing, 2025
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- IEEE WCL **Deep Learning-based Channel Estimation for Double-RIS Aided Massive MIMO System**
 Mengbing Liu, **Xin Li**, Boyu Ning, Chongwen Huang, Sumei Sun, Chau Yuen
IEEE Wireless Communications Letters, 2022
[Robotics & Computer Vision](#)
- IEEE RA-L **Co-planar Parametrization for Stereo-SLAM and Visual-Inertial Odometry**
Xin Li*, Yanyan Li*, Evin Pinar Örneke, Jinlong Lin, Federico Tombari
IEEE Robotics and Automation Letters (RA-L), 2020
 [arXiv] [Code]
- IROS 2020 **Leveraging Planar Regularities for Point Line Visual-Inertial Odometry**
Xin Li*, Yijia He*, Jinlong Lin, Xiao Liu
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2020
- IEEE Sensors Journal **PVI-DSO: Leveraging Planar Regularities for Direct Sparse Visual-Inertial Odometry**
 Bo Xu, **Xin Li**, Jingrong Wang, Chau Yuen, Jiancheng Li
IEEE Sensors Journal, 2023

Experience

- 2024–2025 **Research Assistant**, *Nanyang Technological University, Singapore*
 Advisor: Prof. Chau Yuen
- Pioneered the application of LLMs to solve complex mathematical problems in wireless communications, creating the first comprehensive benchmark (WirelessMathBench) for evaluating LLMs in this domain
 - Developed advanced robotic perception systems integrating deep learning for real-time dense 3D reconstruction
- 2022–2024 **SLAM Algorithm Engineer (Team Lead)**, *Gausium Robotics, Singapore*
- Led a team of 5 engineers in developing a hierarchical visual localization system for large-scale environments, achieving 95% localization accuracy across 100,000+ sq.m. spaces
 - Architected real-time map update modules processing 10+ FPS on embedded systems using compact neural networks
 - Deployed solutions in 1000+ commercial robots across worldwide
- 2021–2022 **Research Assistant**, *Singapore University of Technology and Design, Singapore*
 Advisor: Prof. Chau Yuen
- Developed a coplanar direct visual-inertial odometry system achieving 30% improvement in accuracy over state-of-the-art planar-constrained methods
 - Published findings in IEEE Sensors Journal (Impact Factor: 4.3)
- 2020–2021 **Research Intern**, *Microsoft Research Asia (MSRA), Beijing, China*
 Mentors: Dr. Yang Liu & Dr. Yizhong Zhang
- Engineered a multi-sensor fusion system integrating RGB-D cameras and IMU data for large-scale indoor mapping
 - Generated high-fidelity vectorized maps of 10,000+ sq.m. supermarkets with sub-meter accuracy
- 2019–2020 **Research Intern**, *MEGVII Technology, Beijing, China*
 Mentor: Dr. Yijia He
- Developed a real-time monocular Visual-Inertial Odometry system leveraging heterogeneous features
 - Achieved semi-dense 3D mesh reconstruction at 30+ FPS on mobile devices
 - Co-authored paper accepted at IROS 2020 (top robotics conference)

Technical Skills

Programming C++ (Expert), Python (Expert), CUDA, ROS/ROS2, PyTorch
AI/ML Large Language Models, Deep Learning, Computer Vision, Reinforcement Learning, Transformer Architectures
Robotics SLAM, Visual-Inertial Odometry, 3D Reconstruction, Sensor Fusion, Path Planning

Academic Service

Reviewer

- **AI/ML Conferences:** NeurIPS (2024, 2025), ICLR (2025), ICML (2025), AISTATS (2025)
- **Robotics:** IROS (2021, 2022), ICRA (2022, 2023, 2025), IEEE RA-L
- **Computer Graphics:** SIGGRAPH (2025), ACM Transactions on Graphics (TOG)

Awards & Achievements

2025 NTU Research Scholarship (Full PhD Funding)
2025 4th Place, ICASSP 2025 Indoor Pathloss Radio Map Prediction Challenge

Languages

English (Fluent), Mandarin Chinese (Native)